Hallucinations and Grounded Answering

Understanding unsupported claims in generated responses

Purpose

This short reference note is written for learning, semantic retrieval, and grounded question-answering experiments. It uses original explanatory content and is organized into coherent sections rather than unrelated filler.

1. What Is a Hallucination?

In language-model applications, a hallucination is an output that presents incorrect or unsupported information as though it were true. A response can sound confident and grammatically polished while containing invented details. Hallucinations may arise from gaps in learned information, ambiguous prompts, conflicting context, weak retrieval, or generation behavior that favors plausible continuation over verified evidence.

2. Common Failure Types

Factual errors state something false. Unsupported claims may be plausible but are not established by the supplied sources. Citation errors point to a source that does not support the claim. Omission failures leave out an important qualification. A model can also misread a passage or combine facts from separate documents incorrectly. These failures should be measured separately when possible because they require different fixes.

3. Grounding with Evidence

A grounded-answering system retrieves passages from a known corpus and supplies them to the model. The response should be based on those passages rather than on unverified recollection. Each passage should retain source metadata such as file name, page, and chunk ID. This enables users to inspect evidence and helps developers trace mistakes to extraction, chunking, retrieval, or generation.

4. Abstention and Unsupported Questions

A reliable document assistant should sometimes say that the supplied documents do not contain enough information. This is not a failure when the question is genuinely unsupported. Evaluation should include “trap” questions whose answers are absent from the corpus. A system that answers every question may appear helpful but can be less trustworthy than one that distinguishes known evidence from missing information.

5. Prompting Is Not a Complete Solution

Instructions such as “use only the provided context” can improve behavior, but a model may still infer beyond the evidence. Retrieval can also return irrelevant or incomplete chunks. Grounding therefore needs multiple layers: relevant retrieval, clear evidence formatting, conservative generation instructions, source-aware output, and evaluation. When possible, check whether cited passages actually entail the claims they accompany.

6. A Debugging Workflow

When an answer is wrong, first inspect the extracted source text. If the relevant information was lost during extraction, fix the parser or cleaning step. If it exists but was split awkwardly, review chunk boundaries and overlap. If it exists in a chunk that was not retrieved, examine embeddings, query wording, top-k, filters, and reranking. If the correct evidence was provided but ignored, revise prompt structure or generation settings and test again.

Key Takeaways

Trustworthy RAG is not simply an LLM connected to a vector database. It is an evidence pipeline that must retrieve relevant material, preserve provenance, handle missing evidence, and measure unsupported claims.

End of document • Use source passages as evidence and verify claims during evaluation.